

SALES PERFORMANCE ANALYSIS

Alfido Tech Data Analytics Internship

Student Details

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Internship Role: Data Analytics Intern

Project Title: Sales Performance Analysis

Tools Used: Python, Pandas, NumPy, Matplotlib, Google Colab

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Introduction

Sales performance analysis provides valuable insights into business growth, product profitability, regional performance, and seasonal sales trends. The objective of this project is to analyze sales data from a retail superstore, evaluate key performance indicators (KPIs), identify high-performing and low-performing products, analyze regional sales trends, and provide actionable recommendations to improve business performance.

Dataset Overview

The dataset contains sales transaction records from a retail superstore. It includes information about orders, customers, products, sales values, shipping details, and regional distribution.

Dataset Details

- Number of Records: 9,800
- Number of Columns: 18

Key Attributes

- Order ID
- Order Date
- Ship Date
- Customer Name

- Segment
- Region
- State
- Category
- Sub Category
- Product Name
- Sales

This dataset provides valuable insights into sales performance across product categories, customer segments, and regions.

Data Cleaning and Preparation

The dataset was cleaned and prepared using Python in Google Colab.

The following steps were performed:

- Loaded the dataset and reviewed its structure.
- Checked for missing values.
- Identified 11 missing values in the Postal Code column.
- Replaced missing Postal Code values with 0.
- Converted Order Date and Ship Date columns into datetime format.
- Created Month and Year columns for trend analysis.
- Verified that the dataset was suitable for analysis and visualization.

	0		<class 'pandas.core.frame.DataFrame'>
Row_ID	0		RangeIndex: 9800 entries, 0 to 9799
Order_ID	0		Data columns (total 18 columns):
Order_Date	0		# Column Non-Null Count Dtype
Ship_Date	0		--- -----
Ship_Mode	0		0 Row_ID 9800 non-null int64
Customer_ID	0		1 Order_ID 9800 non-null object
Customer_Name	0		2 Order_Date 9800 non-null object
Segment	0		3 Ship_Date 9800 non-null object
Country	0		4 Ship_Mode 9800 non-null object
City	0		5 Customer_ID 9800 non-null object
State	0		6 Customer_Name 9800 non-null object
Postal_Code	0		7 Segment 9800 non-null object
Region	0		8 Country 9800 non-null object
Product_ID	0		9 City 9800 non-null object
Category	0		10 State 9800 non-null object
Sub_Category	0		11 Postal_Code 9789 non-null float64
Product_Name	0		12 Region 9800 non-null object
Sales	0		13 Product_ID 9800 non-null object
			14 Category 9800 non-null object
			15 Sub_Category 9800 non-null object
			16 Product_Name 9800 non-null object
			17 Sales 9800 non-null float64
			dtypes: float64(2), int64(1), object(15)
			memory usage: 1.3+ MB

Key Performance Indicators (KPIs)

The following KPIs were calculated:

KPI	Value
Total Revenue	2,261,536.78
Average Order Value	230.77
Total Orders	4,922

These metrics provide a clear overview of overall business performance and customer purchasing behavior.

```
total_revenue = df["Sales"].sum()
print("Total Revenue:", total_revenue)

Total Revenue: 2261536.7827000003

avg_order_value = df["Sales"].mean()
print("Average Order Value:", avg_order_value)

Average Order Value: 230.7690594591837

total_orders = df["Order_ID"].nunique()
print("Total Orders:", total_orders)

Total Orders: 4922
```

Category Performance

Sales performance was analyzed across product categories.

Category	Sales
Technology	827,455.87
Furniture	728,658.58
Office Supplies	705,422.33

Technology generated the highest sales revenue, making it the best-performing category.

```
Category
Technology      827455.8730
Furniture        728658.5757
Office Supplies  705422.3340
Name: Sales, dtype: float64
```

Regional Performance

Sales were also analyzed across regions.

Region	Sales
West	710,219.68
East	669,518.73
Central	492,646.91
South	389,151.46

The West region generated the highest revenue, while the South region recorded the lowest sales performance.

```
Region
West      710219.6845
East      669518.7260
Central    492646.9132
South      389151.4590
Name: Sales, dtype: float64
```

Data Visualizations

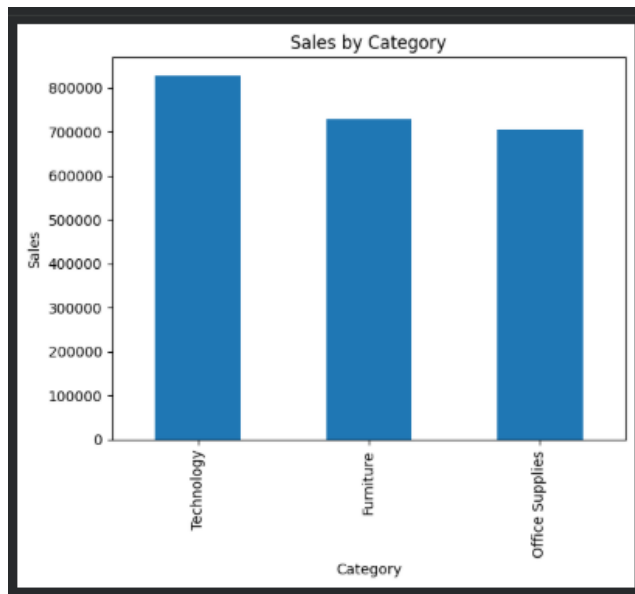


Figure 1: Sales by Category

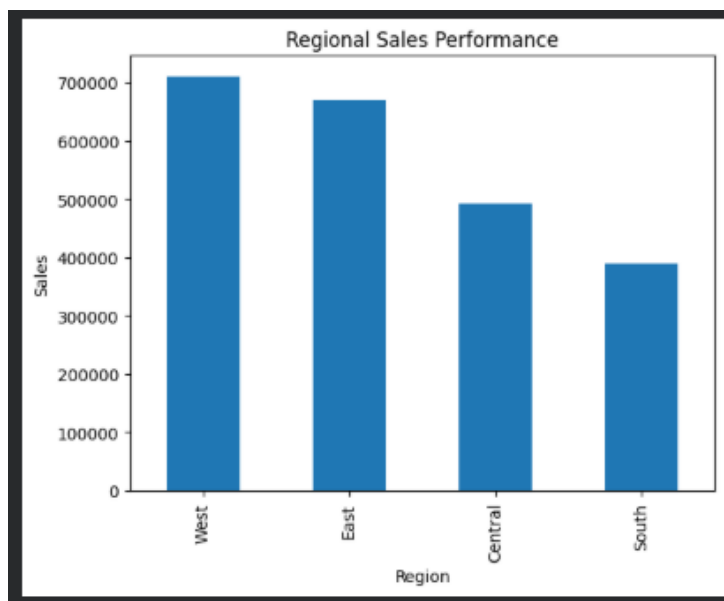


Figure 2: Regional Sales Performance

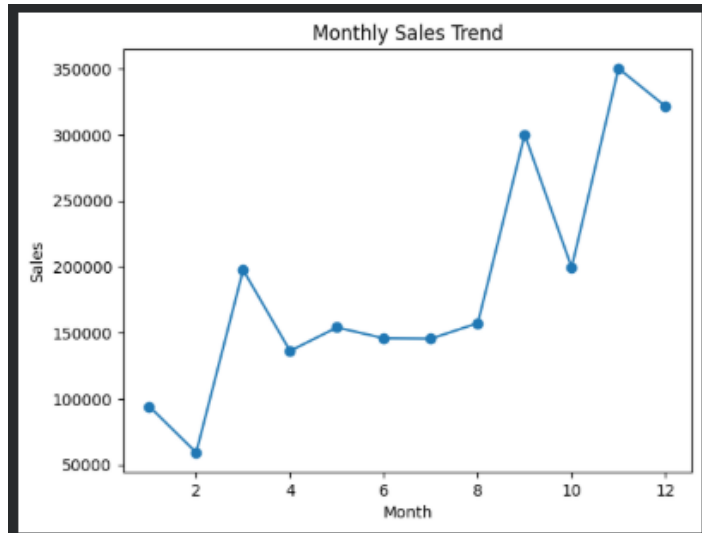


Figure 3: Monthly Sales Trend

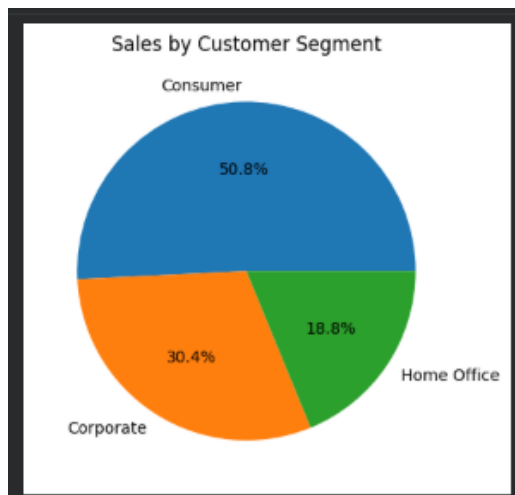


Figure 4: Sales by Customer Segment

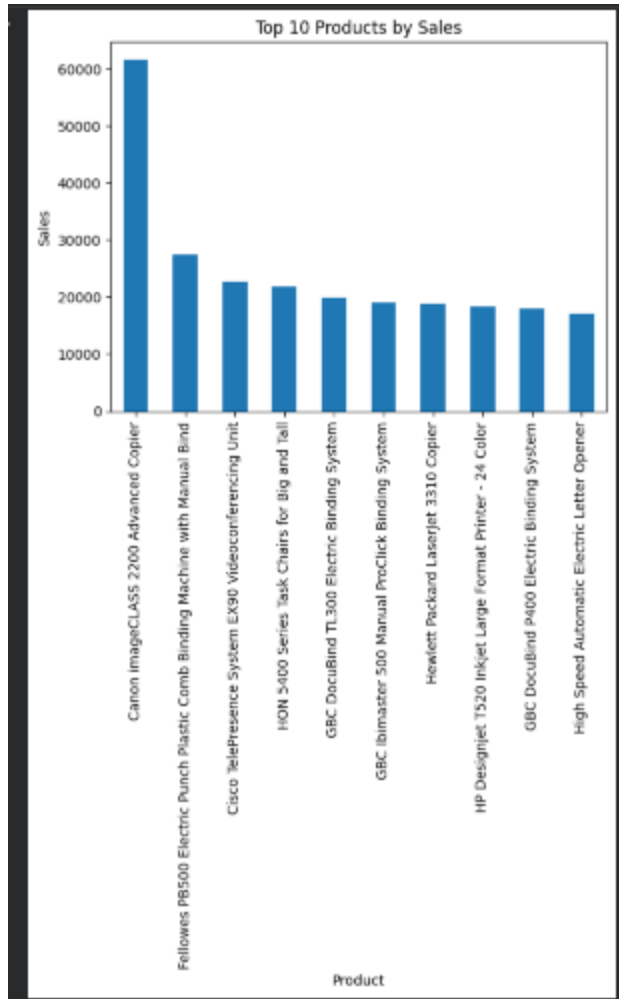


Figure 5: Top 10 Products by Sales

Product Performance Analysis

Top Performing Product

The highest-selling product was:

Canon imageCLASS 2200 Advanced Copier – 61,599.82

This product generated significantly higher revenue compared to other products in the dataset.

Lowest Performing Product

The lowest-selling product was:

Eureka Disposable Bags for Sanitaire Vibra Groomer I Upright Vac – 1.624

This product contributed very little to the overall revenue.

	Sales
Product_Name	
Canon imageCLASS 2200 Advanced Copier	61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.384
Cisco TelePresence System EX90 Videoconferencing Unit	22638.480
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479
GBC Ibimaster 500 Manual ProClick Binding System	19024.500
Hewlett Packard LaserJet 3310 Copier	18839.686
HP Designjet T520 Inkjet Large Format Printer - 24 Color	18374.895
GBC DocuBind P400 Electric Binding System	17965.068
High Speed Automatic Electric Letter Opener	17030.312

dtype: float64

	Sales
Product_Name	
Eureka Disposable Bags for Sanitaire Vibra Groomer I Upright Vac	1.624
Avery 5	5.760
Xerox 20	6.480
Grip Seal Envelopes	7.072
Acme Serrated Blade Letter Opener	7.632
Avery Hi-Liter Pen Style Six-Color Fluorescent Set	7.700
Avery Hi-Liter Comfort Grip Fluorescent Highlighter, Yellow Ink	7.800
Xerox 1989	7.968
4009 Highlighters	8.040
Stockwell Gold Paper Clips	8.096

dtype: float64

Seasonality Analysis

Monthly sales analysis revealed strong seasonal trends.

Highest Sales Months

1. November – 350,161.71
2. December – 321,480.17
3. September – 300,103.41

These months generated the highest revenue and indicate increased customer purchasing activity toward the end of the year.

Sales	
Month	
1	94291.6296
2	59371.1154
3	197573.5872
4	136283.0006
5	154086.7237
6	145837.5233
7	145535.6890
8	157315.9270
9	300103.4117
10	199496.2947
11	350161.7110
12	321480.1695
dtype: float64	

Key Findings

1. The total revenue generated was **2.26 million**.
2. The average order value was **230.77**.
3. Technology was the highest-performing category.
4. The West region generated the highest sales revenue.
5. The South region recorded the lowest sales performance.
6. Canon imageCLASS 2200 Advanced Copier was the best-selling product.

7. November and December recorded the highest monthly sales, indicating strong seasonal demand.
8. Sales performance varied significantly across products and regions.

Recommendations

Based on the analysis, the following recommendations are suggested:

1. Increase inventory and promotional campaigns for high-performing Technology products.
2. Develop targeted sales strategies for the South region to improve regional performance.
3. Promote top-performing products through bundled offers and featured marketing campaigns.
4. Prepare inventory and marketing activities in advance for peak sales periods such as November and December.
5. Review low-performing products and consider discounts, repositioning, or discontinuation if they do not contribute meaningfully to revenue.

Conclusion

The sales performance analysis provided valuable insights into category performance, regional sales distribution, product profitability, and seasonal trends. The findings indicate that focusing on high-performing categories, improving underperforming regions, and preparing for peak sales periods can help enhance business growth and profitability. These insights support data-driven decision-making and contribute to improved sales performance.

EXECUTIVE SUMMARY

This project focused on analyzing sales performance across product categories, regions, and time periods using retail sales data. The analysis revealed total revenue of approximately 2.26 million and an average order value of 230.77. Technology emerged as the top-performing category, while the West region generated the highest sales revenue. Seasonal trends showed peak sales during November and December, indicating strong year-end demand. The study recommended focusing on high-performing products, improving sales strategies in underperforming regions, and planning inventory and marketing activities around peak sales periods to maximize revenue growth.